



Universidade do Minho

Bases de Dados

Licenciatura em Ciências da Computação

Ano Letivo de 2025/2026

2º Semestre

PL10

Topic: SQL – Indexes, Procedures, Triggers, Functions, Transactions

Solve the following exercises using the Sakila database.

Exercises

1. Create an index on the `payment_date` column of the `payment` table.
2. Create a prepared statement that returns the names of customers for each store. Execute the prepared statement for store 1.
3. Create a procedure called `filmes_por_categoria` that receives a category ID and returns the titles of the films in that category. Execute the procedure for a category of your choice.
4. Create a procedure that receives a `customer_id` and returns the date of the last payment.
5. Create a function `total_filmes_ator` that receives an actor ID and returns the number of films for that actor. Execute the function for an actor of your choice.
6. Create a trigger that, after inserting a new record into the `payment` table, automatically updates the `total_spent` field in the `customer` table by adding the payment amount (`amount`) to the customer's existing total.
7. Create a transaction that updates a customer to `active = 1` and inserts a payment.
8. Create a transaction that updates a customer but uses `ROLLBACK`.